

Outlook / Calendar Integration Guide

WilliamOS (Mini-jarv) — setup, usage, and limitations

Overview

The Outlook/calendar integration pulls **calendar events** and **unread emails** from Microsoft Outlook into your **Inbox** as signals. You can then apply AI suggestions to create tasks, assets, and more.

Stack: Microsoft Graph API → FastAPI (outlook_service.py) → Inbox (capture_inbox_entry)

What's Already Built

You do **not** need to write integration code — the full path is wired:

Layer	Status	Location
OAuth flow	Done	app/services/outlook_service.py
Token refresh	Done	Auto-refreshes before sync
Calendar fetch	Done	Next 7 days, max 15 events
Email fetch	Done	Last 8 messages, unread only
API routes	Done	app/api/routes/integrations.py
UI	Done	/integrations + /integrations/callback
DB table	Done	user_integrations in migration

OAuth flow

1. **Integrasjoner** → **Koble til** → POST /integrations/outlook/connect
2. Redirect to Microsoft login
3. Microsoft redirects to /integrations/callback?code=...&state=...
4. Callback calls POST /integrations/outlook/complete
5. Tokens stored in user_integrations

Sync flow

1. **Synk til Inbox** → `POST /integrations/outlook/sync`
 2. Fetches calendar + unread mail via Microsoft Graph
 3. Creates Inbox lines like:
 - Outlook kalender: Møte (2026-08-18 10:00)
 - Outlook e-post: Forsikring fornyes
 4. Each line gets LLM/rule-based suggestions — you apply them manually in Inbox
-

What You Need to Do

1. Run the Supabase migration

In Supabase SQL editor, run:

```
migrations/2026-08-17_finance_health_integrations.sql
```

This creates the `user_integrations` table where OAuth tokens are stored.

2. Register a free Azure app (~15 min)

Go to: <https://portal.azure.com> → **App registrations**

1. New registration

2. Redirect URI (Web):

- Local: `http://localhost:3000/integrations/callback`
- iPhone via ngrok: `https://YOUR-SUBDOMAIN.ngrok-free.app/integrations/callback`

3. Certificates & secrets → create a **Client secret** (copy immediately)

4. API permissions → Microsoft Graph → **Delegated**:

- `Calendars.Read`
- `Mail.Read`
- `User.Read`
- `offline_access` (for refresh token)

5. **Grant admin consent** if your tenant requires it

Note: This is free — no paid Azure subscription needed.

3. Set environment variables

Add to your root `.env`:

```
MICROSOFT_CLIENT_ID=your-app-client-id
MICROSOFT_CLIENT_SECRET=your-client-secret
MICROSOFT_REDIRECT_URI=http://localhost:3000/integrations/callback
FRONTEND_URL=http://localhost:3000
```

Critical: MICROSOFT_REDIRECT_URI must match Azure **character for character**.

4. Restart the backend

FastAPI reads env at startup. Without restart, the Integrations page shows "Sett MICROSOFT_CLIENT_ID..." and the connect button stays disabled.

5. Connect in the app

1. Start backend (:8000) + frontend (:3000)
2. Log in to Mini-jarv
3. Go to **Mer** → **Integrasjoner**
4. Click **Koble til** on Outlook → Microsoft login → approve
5. Click **Synk til Inbox** → check **Inbox**

6. Use a Microsoft account with calendar/email

Personal @outlook.com / @hotmail.com or work Microsoft 365 both work via the /common OAuth endpoint.

Required Scopes

`offline_access Calendars.Read Mail.Read User.Read`

Defined in `app/services/outlook_service.py`.

Known Limitations (Today)

Limitation	Detail
Manual sync only	No cron, no sync-on-login
Duplicates	Re-syncing creates duplicate Inbox items
Unread email only	Read mail is skipped
Calendar window	7 days ahead, max 15 events
No calendar UI	Events only appear as Inbox text lines
Gmail/iCloud	Not supported — Microsoft Graph only
Work/school tenants	May need admin consent for Mail.Read

Limitation	Detail
No auto-tasks	Sync creates Inbox signals; tasks are not auto-created

Troubleshooting

Problem	Solution
"Sett MICROSOFT_CLIENT_ID..." in UI	All three env vars missing, or backend not restarted
Redirect URI mismatch	Azure URI must equal MICROSOFT_REDIRECT_URI exactly
"Ugyldig OAuth-state"	Click Koble til again (state stored in user_integrations.metadata)
Koble til button disabled	configured: false — check env vars
ngrok on iPhone	Update both Azure redirect URI and MICROSOFT_REDIRECT_URI to ngrok URL
No inbox after sync	Calendar may be empty next 7 days, or all mail is read
Token expired / error status	Disconnect and reconnect Outlook

API Reference (Optional)

All endpoints require authenticated user (Bearer JWT).

GET /integrations	List integration statuses
POST /integrations/outlook/connect	Start OAuth, returns auth_url
POST /integrations/outlook/complete	Complete OAuth with code + state
POST /integrations/outlook/sync	Sync calendar + mail to Inbox
POST /integrations/outlook/disconnect	Disconnect and clear tokens

Pre-Flight Checklist

- Migration 2026-08-17_finance_health_integrations.sql run in Supabase
- Azure app registered with correct redirect URI
- MICROSOFT_CLIENT_ID, MICROSOFT_CLIENT_SECRET, MICROSOFT_REDIRECT_URI in .env

- Backend restarted after env changes
 - Logged into Mini-jarv
 - Outlook account has upcoming calendar events and/or unread mail
-

Future Improvements (Not Built Yet)

- Dedup on sync (avoid duplicate Inbox rows)
 - Calendar event → auto-suggest/create task
 - Email keyword parsing (forsikring, faktura → asset/task)
 - Auto-sync on login or scheduled daily sync
 - Dedicated calendar view in the app
-

Generated for WilliamOS / Mini-jarv — see also docs/GETTING-STARTED.md and MVP-FOCUS.md